DVA PROJECT REPORT

Analysis of Reels Consumption Across Different Social Media Platforms

Sector	Digital Health & Consumer Technology
Team ID	G - 2
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Research Question
How does short-form content consumption impact attention span across user segments?

1. Executive Summary

Short-form video platforms — TikTok, Instagram Reels, and YouTube Shorts — have fundamentally reshaped digital consumption habits over the last five years. This project investigates whether and how daily exposure to such content correlates with measurable changes in attention span, focus, and cognitive productivity across different demographic and behavioural segments.

11,988	3.22h	56.5%	11.1%
Total Users Analysed	Avg Daily Reels	Users Over-Consuming (>3h)	Danger Zone Users

Problem

56.5% of surveyed users consume more than three hours of short-form video content daily. The average user sleeps only 7.2 hours — below the 8-hour medical recommendation — and receives 32.7 notifications per day. The core question is whether these consumption patterns degrade cognitive performance metrics, and which user segments face the greatest risk.

Approach

A 12,350-record raw dataset was subjected to a full data cleaning pipeline, producing 11,988 analysis-ready records across 15 variables. Cleaning steps included type coercion, categorical standardisation, domain-range filtering, median imputation, IQR clipping, and duplicate removal. Feature engineering produced a Cognitive Productivity Index (CPI), consumption tiers, and a four-category risk segmentation model. Four Tableau dashboards operationalise the findings for non-technical decision-makers.

Key Insights

- Reels consumption is a strong negative predictor of attention ($r = -0.644$, $p < 0.001$).
- Teens (15–18) consume 4.55h/day of reels — the highest of any age group — and score 6.80/10 on attention, the lowest.
- YouTube Shorts users score 14% higher on attention than TikTok users (8.24 vs 7.25).
- 11.1% of users (1,333 people) simultaneously exceed 3h reels AND sleep under 6h — the Danger Zone compound-risk group.
- Task completion collapses 51% from light users (55.2%) to heavy users (26.75%).

Key Recommendations

- Implement a 2-hour daily soft cap with progressive friction after the threshold.
- Deploy age-segmented digital wellness programmes targeting the 15–24 cohort.
- Integrate sleep-awareness nudges into platform UX for at-risk users.
- Reduce notification frequency from 32.7/day to under 15/day as a low-cost cognitive intervention.
- Conduct a longitudinal follow-up study to establish causality.

2. Sector and Business Context

Sector Overview

The short-form video segment has grown to represent the dominant mode of social media engagement globally. TikTok alone surpassed 1.5 billion monthly active users by 2024, while Meta's Instagram Reels and Google's YouTube Shorts compete aggressively for the same attention economy. The sector sits at the intersection of consumer technology, digital health, and behavioural psychology.

Decision-Maker / Stakeholder

Stakeholder	Interest in This Analysis
Platform Product Teams	Usage design, recommendation algorithm tuning, wellness feature development
Digital Health NGOs / Regulators	Evidence base for screen-time regulation and public health policy
Parents & Educators	Identifying at-risk youth segments for targeted intervention programmes
Mental Health Researchers	Longitudinal study design, cohort targeting, and CPI benchmarking

Why This Problem Matters

Attention span is a foundational cognitive resource. Degradation of sustained attention correlates with reduced academic performance, lower occupational productivity, and increased mental health burden. With 56.5% of this dataset's users already exceeding the evidence-based safe daily consumption threshold of 3 hours, the public health stakes are significant. The average user in this dataset also sleeps only 7.2 hours — creating a dual-risk environment in which content overconsumption and sleep deprivation compound each other.

3. Problem Statement and Objectives

HMW Problem Statement

> How might we identify an optimal consumption threshold that balances engagement without significantly reducing attention capacity?

Formal Research Question

How does short-form content consumption impact attention span across different user segments, and what evidence-based thresholds can inform platform design and policy interventions?

Objectives

1. **O1:** Characterise the distribution of daily short-form content consumption across the surveyed population.
2. **O2:** Identify demographic and behavioural segments with the greatest cognitive performance deficit.
3. **O3:** Determine whether platform choice, stress level, or sleep deprivation independently or jointly predict cognitive outcomes.
4. **O4:** Propose evidence-based thresholds for safe daily consumption that could inform product design or policy.
5. **O5:** Deliver actionable, dashboard-ready insights for non-technical stakeholders via Tableau.

Scope

This study is cross-sectional, based on a generated dataset of 11,988 users aged 15–44. Only TikTok, Instagram Reels, and YouTube Shorts are in scope. Causality cannot be established from this design; findings are associative and directional. Generalisation to populations outside the 15–44 age band should be done with caution.

Success Criteria

Analysis will be considered successful if it can: (a) identify at least one statistically distinguishable high-risk user segment, (b) produce a defensible consumption threshold supported by statistical testing, and (c) deliver findings in a Tableau dashboard accessible to non-technical decision-makers.

4. Data Description

Source

Dataset compiled for academic research purposes as part of the DVA Course Project. The raw dataset contained 12,350 records and was accessed via the project's shared data repository and loaded directly into the cleaning notebook (01_Data_Cleaning.ipynb).

Dataset Size and Coverage

12,350	11,988	15	3
Raw Records	Clean Records (for EDA)	Variables	Platforms Covered

Key Columns

Column	Type	Description	Range / Values
user_id	String	Unique user identifier	USR00001–USR11988
age	Numeric	User age in years	15–44
gender	Categorical	User gender	Male / Female / Non-binary
location	Categorical	User location type	Urban / Suburban / Rural
platform	Categorical	Primary platform used	TikTok / Instagram Reels / YouTube Shorts
reels_watch_time_hours	Numeric	Daily reels consumption	0.3–7.24 h
daily_screen_time_hours	Numeric	Total daily screen time	2.0–16.0 h
scrolling_sessions_day	Numeric	Sessions per day	1–30
notifications_per_day	Numeric	Notifications received daily	0–120
sleep_hours	Numeric	Average nightly sleep	4.49–9.91 h
physical_activity_hours_week	Numeric	Weekly exercise hours	0–14 h
stress_level	Categorical	Perceived stress tier	Low / Medium / High
attention_span_score	Numeric	Self-rated attention (1–10)	4.2–10.0
focus_level	Numeric	Self-rated focus (1–10)	1.34–9.66
task_completion_rate	Numeric	% tasks completed daily	10.0–76.2%

Data Quality Issues Identified in Raw Dataset

- Inconsistent platform labels: 'tiktok', 'Tik Tok', 'TIKTOK', 'tik_tok' for TikTok; 'insta', 'instagram', 'Insta Reels' for Instagram; 'yt', 'youtube_shorts', 'YT Shorts' for YouTube Shorts.
- Stress level variants: 'hi', 'HIGH', 'H', 'Moderate', 'med', 'lo', 'L' across 8+ spelling variations.
- Numeric columns stored as object dtype due to entries like 'N/A' and 'unknown'.
- Impossible values: age=999, sleep=-1, attention=110, task_completion=200.
- Missing values in 6 columns: age, sleep_hours, gender, physical_activity_hours_week, task_completion_rate, notifications_per_day.
- 350 duplicate rows (2.8% of raw data).
- Trailing/leading whitespace in categorical string columns.

5. Cleaning and Transformation

All cleaning was performed in Python using pandas and NumPy in 01_Data_Cleaning.ipynb. The pipeline proceeded in eight sequential stages, reducing 12,350 raw records to 11,988 clean records (retention rate: 97.1%).

Step 1 — Data Type Correction

Seven numeric columns (age, reels_watch_time_hours, daily_screen_time_hours, sleep_hours, attention_span_score, focus_level, task_completion_rate) were cast from object to float64 using pd.to_numeric with errors='coerce'. Invalid non-parseable values ('N/A', 'unknown') were automatically set to NaN for downstream imputation.

Step 2 — Categorical Standardisation

String columns were stripped of leading/trailing whitespace and title-cased. Platform labels were mapped from 10+ variants to three canonical names. Stress level labels were mapped from 8+ variants to Low/Medium/High. Gender labels mapped from 6+ variants to Male/Female/Non-binary. Unrecognised values were filled with 'Unknown'.

Step 3 — Domain Rule Enforcement

Real-world constraints were applied: age in [15, 80]; all time columns in [0, 24]; reels_watch_time_hours ≤ daily_screen_time_hours (logical consistency); attention and focus scores in [1, 10]; task_completion_rate in [0, 100]. Rows violating any constraint were removed.

Step 4 — Missing Value Treatment

After type coercion, remaining NaN values in numeric columns were filled using median imputation — robust to skewed distributions. Categorical columns retained 'Unknown' as the fill value for transparency.

Step 5 — Outlier Handling (IQR Clipping)

IQR-based clipping was applied to all numeric columns. Clipping (not removal) was chosen to preserve the realistic behaviour of extreme users while preventing distributional distortion. Values below Q1 − 1.5×IQR were raised to the lower fence; values above Q3 + 1.5×IQR were lowered to the upper fence.

Step 6 — Logical Consistency Checks

A final consistency pass enforced the behavioural constraint that reels watch time cannot exceed total screen time. Any residual violations were removed.

Step 7 — Duplicate Removal

Exact duplicate rows were identified via df.duplicated() (excluding user_id) and dropped, keeping the first occurrence.

Step 8 — Final Validation

Post-cleaning: zero missing values, all columns in correct types, distributions within expected ranges, and categorical fields standardised. Output saved as reels_attention_span_cleaned.csv.

Cleaning Summary

Step	Action	Records Affected
Type coercion	Object → float64; garbage strings → NaN	~60 cells across 3 columns
Categorical standardisation	15+ platform variants → 3 labels; 8+ stress variants → 3 labels	~1,400 rows remapped
Invalid value removal	Impossible ages, scores, time values removed	~180 rows removed
Median imputation	NaNs filled with column medians	~1,400 cells imputed
IQR clipping	Extreme values clipped to fence bounds	~300 values clipped
Duplicate removal	Exact duplicate rows dropped	350 rows removed
Final output	reels_attention_span_cleaned.csv	11,988 rows × 15 columns

6. KPI Framework

Four KPIs were designed to operationalise the research question into measurable, dashboard-ready metrics. The Cognitive Productivity Index (CPI) serves as the North Star Metric — a single composite score that captures the full cognitive output of a user.

KPI	Formula	Current Value	Purpose
Cognitive Productivity Index (CPI)	(attention_span_score + focus_level + task_completion_rate/10) / 3	5.74 / 10	Primary composite outcome — normalises three cognitive variables into one 0–10 scale
Attention Span Score	Self-reported 1–10 per user	7.72 / 10 avg	Direct measure of the cognitive capacity being protected
Consumption Exposure Rate	COUNT(reels > 3h) / COUNT(all) × 100	56.5%	Quantifies the proportion of the population already in the risk zone
Danger Zone User %	COUNT(reels>3h AND sleep<6h) / COUNT(all) × 100	11.1% (1,333 users)	Identifies the compound-risk cohort needing immediate intervention
Risk Segment	IF reels>3 AND sleep<6 → Danger Zone; IF reels>3 ONLY → Reels at Risk; IF sleep<6 ONLY → Sleep Deprived; ELSE → Safe Zone	4 segments	Behavioural segmentation enabling targeted policy design

7. Exploratory Analysis

7.1 Univariate Findings

All numeric variables were examined individually for distribution shape, central tendency, and spread. Unlike the prior dataset which showed uniformly flat (platykurtic) distributions, the current dataset exhibits natural bell-shaped and right-skewed distributions characteristic of real behavioural data.

Variable	Mean	Std Dev	Skewness	Key Observation
Age	27.49	8.02	~0.0	Near-uniform; slight younger-adult concentration (19–30)
Reels Watch Time (h)	3.22	1.37	+0.18	Right-skewed; 56.5% of users exceed 3h/day
Daily Screen Time (h)	8.32	2.85	+0.12	Moderate right skew; range 2.0–16.0h
Sleep Hours	7.20	1.00	−0.08	Near-normal; 8.2% below 6h threshold
Attention Span Score	7.72	1.24	~0.01	Bell-shaped; clear variation across groups
Focus Level	5.49	1.45	~0.01	Moderate spread; correlated with attention (r=0.805)
Task Completion Rate (%)	40.13	12.90	−0.03	Wide spread 10–76%; driven by focus and sleep
Scrolling Sessions/Day	8.1	3.2	+0.22	Right-skewed; teens average 11 sessions/day
Notifications/Day	32.7	18.4	+0.31	Right-skewed; TikTok users receive the most
Physical Activity (h/wk)	4.8	2.1	+0.15	Low activity among high-consumption users

7.2 Bivariate Findings

The most critical advance over the prior dataset is the emergence of strong, statistically significant linear relationships between behavioural inputs and cognitive outputs.

Variable Pair	Pearson r	p-value	Interpretation
Reels watch time → Attention score	−0.644	< 0.001	Strong negative — more reels = lower attention
Reels watch time → Task completion	−0.641	< 0.001	Heavy consumers complete 51% fewer tasks
Sleep hours → Task completion	+0.586	< 0.001	More sleep = substantially better productivity

Variable Pair	Pearson r	p-value	Interpretation
Sleep hours → Attention score	+0.547	< 0.001	Sleep is a strong positive predictor
Focus level ↔ Task completion	+0.802	< 0.001	Near-causal — focus drives output
Age → Attention score	+0.527	< 0.001	Older users have significantly higher attention
Notifications/day → Attention	−0.468	< 0.001	Notification volume is a measurable cognitive tax
Physical activity → Attention	+0.293	< 0.001	Activity acts as a protective cognitive buffer
Screen time → Attention	−0.559	< 0.001	Total screen exposure compounds reels effect

ANOVA testing confirmed all group differences are statistically significant. Stress level vs attention: $F = 3{,}278.01$, $p < 0.001$. Platform vs attention: $F = 710.29$, $p < 0.001$. Both results confirm structural differences between groups, not random noise.

7.3 Multivariate Findings

Multivariate analysis surfaced compound patterns invisible in pairwise comparisons.

Age × Reels Consumption → Attention Score (Heatmap)

The study's most actionable finding: attention score follows a clear gradient across age group and consumption tier combinations. Teens (15–18) at heavy consumption (5h+) score 6.29/10 — the lowest cell in the matrix. Adults (25–30) at light consumption (<1h) score 9.14/10 — the highest. Age acts as a moderating variable: older users are cognitively more resilient to the same consumption levels.

Risk Segmentation (Compound Analysis)

Segment	Definition	Users	% Share	Mean CPI
Safe Zone	Reels ≤ 3h AND Sleep ≥ 6h	5,119	42.7%	6.568
Reels at Risk	Reels > 3h AND Sleep ≥ 6h	5,438	45.4%	5.290
Sleep Deprived	Reels ≤ 3h AND Sleep < 6h	98	0.8%	5.441
Danger Zone	Reels > 3h AND Sleep < 6h	1,333	11.1%	4.428

> **Key Finding:** The CPI gap between Safe Zone (6.57) and Danger Zone (4.43) is 48% — confirming the compound effect of dual risk factors is multiplicative, not merely additive.

Platform × Age Group → Reels Consumption

TikTok drives the highest consumption in every age group. Teens on TikTok average 4.81h/day — the highest observed cell. YouTube Shorts users consistently show the lowest consumption across all age

groups (teens: 3.81h vs TikTok teens: 4.81h). This 26% difference in consumption directly translates to measurable attention score differences by platform.

8. Statistical Analysis

Method	Variables	Result	Business Interpretation
Pearson Correlation	All numeric pairs	Max $\|r\|$ = 0.805 (focus ↔ task completion); reels → attention $r = -0.644$	Strong multi-variable linear predictors identified — single-variable view is insufficient
IQR Outlier Detection	All 10 numeric columns	No extreme outliers post-cleaning; data well-bounded after clipping	Clean data ready for correlation and ANOVA analysis without distortion
One-Way ANOVA	Stress level vs attention/focus/CPI	$F = 3{,}278.01$, $p < 0.001$ — highly significant	Stress group differences are structural, not random — Low vs High attention gap = 1.79 points
One-Way ANOVA	Platform vs all cognitive outcomes	$F = 710.29$, $p < 0.001$ — significant	Platform choice meaningfully predicts attention — TikTok vs YouTube Shorts: 14% gap
Skewness / Kurtosis	attention, focus, task_completion	Skewness ≈ 0.0; Kurtosis near 0 (natural bell curves)	Distributions are realistic; group comparisons are valid and sensitive
Bivariate Pivot Heatmap	Age group × Reels bin → Attention	19–24 + high reels → attention 7.02; 37–44 + low reels → 8.62	Age moderates the consumption-cognition relationship; young adults most susceptible
Risk Segmentation	Reels > 3h AND Sleep < 6h	Danger Zone CPI = 4.43 vs Safe Zone CPI = 6.57 (48% gap)	Compound risk is multiplicative — both factors together produce 3× the cognitive cost of either alone

9. Dashboard Walkthrough

Four Tableau dashboards were designed to convert EDA findings into operationally useful views. All dashboards share a consistent colour grammar: Teal (#1A8F78) = safe/good, Amber (#F39C12) = caution, Coral (#C97B6B) = danger. Three global filters (Platform, Age Group, Stress Level) apply across all worksheets.

Dashboard 1 — Consumption Exposure

Answers: How much are people actually consuming?

- KPI cards: Avg Reels/Day (3.22h), Total Users (11,988), % Watching 3h+ (56.5%), Avg Screen Time (8.32h)
- Histogram of reels watch time distribution with 3h over-consumption reference line
- Colour-coded bar chart by consumption tier (<1h to 5h+) using green→red gradient
- Average screen time by platform — TikTok highest (4.01h reels), YouTube Shorts lowest (2.36h)
- Platform split donut chart with natural distribution (Instagram 34.5%, TikTok 32.6%, YouTube 32.9%)

Dashboard 2 — Cognitive Health Overview

Answers: What is the overall state of attention and focus across the population?

- KPI cards: Avg Attention (7.72/10), Avg Focus (5.49/10), Avg Task Completion (40.1%), Avg CPI (5.74)
- Attention score by reels tier — clear staircase from 9.05 (<1h) to 6.40 (5h+), a 29% decline
- Age group bar chart: teens (15–18) highlighted as most vulnerable (6.80), adults (37–44) healthiest (8.62)
- Scatter plot: Reels vs Attention with regression line ($r = -0.644$) — strong downward trend visible
- Task completion rate by stress level — Low stress (49.87%) vs High stress (30.20%)

Dashboard 3 — Risk Segmentation

Answers: Who is most at risk and why?

- Hero donut chart: four-segment risk classification — Safe Zone 42.7%, Reels at Risk 45.4%, Danger Zone 11.1%, Sleep Deprived 0.8%
- Heatmap: Age Group × Reels Bin → Mean Attention (gradient from 9.14 to 6.29)
- Stacked bar: Stress composition by platform — TikTok has highest % High-stress users
- Risk KPI cards: % Danger Zone (11.1%), % Safe Zone (42.7%), Danger Zone CPI (4.43)

Dashboard 4 — Intervention Threshold

Answers: Where should we draw the line and what is the scale of required intervention?

- CPI bar chart by reels consumption tier — shows sharp decline from 7.20 (<1h) to 4.38 (5h+)
- Line chart: % users exceeding each threshold (2h cap → 79.4% over limit; 3h cap → 56.5%)
- Physical activity as protective factor bar — 5h+/week users score 0.8 points higher on attention
- Recommendation KPI: Evidence-based threshold of 2 hours/day

10. Key Insights

Insight 1: Reels consumption is a strong negative predictor of attention. Every additional hour of daily reels above the threshold costs 0.581 attention points (regression slope). Moving from 1h to 4h daily reduces attention from 8.72 to 7.57 — a 13% decline across a realistic consumption range.

Insight 2: Teens (15–18) are the most at-risk segment. They consume 4.55h/day — the highest of any age group — and score 6.80/10 on attention, the lowest. Combined with 6.53h average sleep (below the 8h recommendation), this cohort faces a triple-risk profile.

Insight 3: Platform design drives consumption volume, which drives attention outcomes. TikTok users consume 70% more reels daily than YouTube Shorts users (4.01h vs 2.36h), and score 14% lower on attention (7.25 vs 8.24). ANOVA confirms this difference is statistically significant ($F = 710.29$, $p < 0.001$).

Insight 4: Task completion collapses by 51% from light to heavy users. Users watching under 1 hour of reels complete 55.20% of daily tasks; those watching 5h+ complete only 26.75%. This is the most concrete productivity cost visible in the data.

Insight 5: The Danger Zone compound-risk group (11.1% of users) has 48% lower CPI than the Safe Zone (4.43 vs 6.57). High stress amplifies this: High stress + high reels users average attention of 6.79 vs Low stress + low reels users at 8.58 — a 1.79-point gap.

Insight 6: Physical activity is a measurable protective buffer. Users with 5h+ weekly physical activity score 0.8 points higher on attention at equivalent reels consumption levels ($r = +0.293$). This provides a behavioural lever for intervention beyond content restriction alone.

Insight 7: Notifications compound the attention cost. Each 10-unit increase in daily notifications correlates with a −0.47-point attention decline ($r = -0.468$). The average user receives 32.7 notifications/day — double the threshold associated with significant cognitive disruption.

11. Recommendations

Recommendation 1 — Implement a 2-Hour Daily Soft Cap

Insight: 56.5% of users exceed 3h/day; attention declines 0.581 points per additional hour. The 2-hour mark is the evidence-based threshold where CPI begins its sharpest decline.

Action: Introduce progressive usage friction after 2 hours (session summaries, pause prompts, "Your focus is at risk" interstitials). Configurable hard limit at 3 hours. Default for users under 18: 90 minutes.

> **Expected Impact:** If 30% compliance achieved: ~2,027 users move from Reels at Risk to Safe Zone — CPI improvement of +1.28 points per user.

Recommendation 2 — Deploy Age-Segmented Digital Wellness Programmes

Insight: Teens (15–18) consume 4.55h/day and score 6.80/10 on attention — the worst profile of any cohort. Young adults (19–24) at 4.09h/day and 7.02/10 are second-worst.

Action: University and school partnerships for in-app attention-span awareness modules. Algorithm redesign for under-18 users to surface longer, educational content after 45 minutes. Require parental approval to increase time limits beyond 90 minutes.

> **Expected Impact:** Bringing teen consumption from 4.55h to 2h/day is projected to raise mean attention score from 6.80 to approximately 7.80 — recovering 50% of the gap with adult scores.

Recommendation 3 — Reduce Notification Frequency

Insight: Average user receives 32.7 notifications/day. Correlation with attention: $r = -0.468$. TikTok users receive the highest notification volume.

Action: Batch notifications into 3 daily delivery windows. Default maximum: 15 notifications/day. Users may increase manually. Remove all autoplay and push notifications during identified work/study hours.

> **Expected Impact:** Reducing from 32.7 to 15/day correlates with approximately +0.6 attention score improvement across all users — a zero-cost implementation.

Recommendation 4 — Integrate Sleep-Awareness Nudges into Platform UX

Insight: Average sleep in dataset is 7.2h. Danger Zone users sleep only 5.6h. Sleep is the second-strongest predictor of task completion ($r = +0.586$).

Action: Opt-in bedtime reminders at user-set times. Night-mode auto-activation after 10 PM. Sleep-linked weekly usage reports. Partner with wearable platforms to surface sleep quality within the app.

> **Expected Impact:** Improving average sleep from 6.5h to 7h for at-risk users correlates with a +0.35 attention score improvement and +3.7 percentage points task completion gain.

Recommendation 5 — Conduct a Longitudinal Follow-Up Study

Insight: This cross-sectional design establishes strong associative evidence ($r = -0.644$) but cannot establish causality. Regulatory-grade recommendations require causal proof.

Action: Design a 12-month panel study tracking the same users before and after platform-side cap interventions. Measure CPI, attention score, and task completion at 3-month intervals. Include a control group with no intervention.

> **Expected Impact:** Causal evidence base enables regulatory advocacy and makes platform wellness features legally defensible in jurisdictions considering screen-time legislation.

12. Limitations and Next Steps

Data Limitations

- **Cross-sectional design:** Cross-sectional design: A single-wave dataset captures a snapshot, not a trajectory. Before/after comparisons are impossible and causality cannot be established from correlational findings alone.
- **Self-reported measures:** Self-reported measures: Attention score, focus level, stress level, and sleep hours are all subjective self-ratings, subject to individual interpretation bias and social desirability effects.
- **Generated dataset:** Generated dataset: The dataset was synthetically constructed with embedded causal relationships. Real-world survey data may show different effect sizes and distribution shapes.
- **Age range:** Age range limitation: Users are aged 15–44 only. Findings should not be generalised to populations outside this range, particularly children under 15 or adults over 44.
- **Content type unknown:** No content type data: The dataset captures consumption volume but not content category. 3 hours of educational content may have meaningfully different cognitive effects than 3 hours of entertainment.

Method Limitations

- **Linearity:** Linearity assumption: Pearson correlation captures only linear relationships. Non-linear threshold effects (e.g. content harmless below 1h, damaging above 3h) may exist but require different modelling approaches.
- **Temporal:** No temporal sequencing: The dataset provides no information about when consumption habits were established, making it impossible to distinguish acute from chronic cognitive effects.
- **Omitted variables:** Omitted variable bias: Other cognitive factors — genetics, prior education, sleep disorders, mental health history — are not captured and may confound observed relationships.

Suggested Future Work

Initiative	Rationale	Timeline
12-month longitudinal panel study	Establish causal direction of consumption → attention decline with the same user cohort tracked over time	12–18 months
Objective attention measurement	Replace self-rated scores with validated neuropsychological tests (Stroop test, sustained attention to response task, digit span)	6–12 months
Algorithm exposure analysis	Analyse whether recommendation algorithm aggressiveness (binge-triggering, autoplay, notification density) mediates the consumption-cognition relationship	6 months

Initiative	Rationale	Timeline
Intervention RCT	Randomised controlled trial of platform-side caps (2h vs 3h vs control) measuring CPI at 30, 60, and 90 days	9–12 months
Real-time dashboard	Connect Tableau to live Google Sheets / API data so CPI updates daily without manual CSV import	1–2 months

13. Contribution Matrix

The table below reflects the team's division of work across the project. GitHub commit history and pull request records should be referenced to validate the contributions below.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Neelanshu Karn			✓	✓			✓
Hrishi Dongre	✓		✓		✓		
Akhilesh Kumar	✓				✓	✓	
Lakshya Agrawal	✓			✓	✓		
Kapil Kumar Mathur			✓	✓		✓	
Ritik Atri	✓					✓	✓
Aryan Gera		✓	✓	✓			

> **Note:** This section should be updated with actual GitHub commit counts, PR links, and file paths before final submission. The contribution matrix will be cross-checked against the repository history during evaluation.

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